NVIDIA

Building AI Agents with Multimodal Models

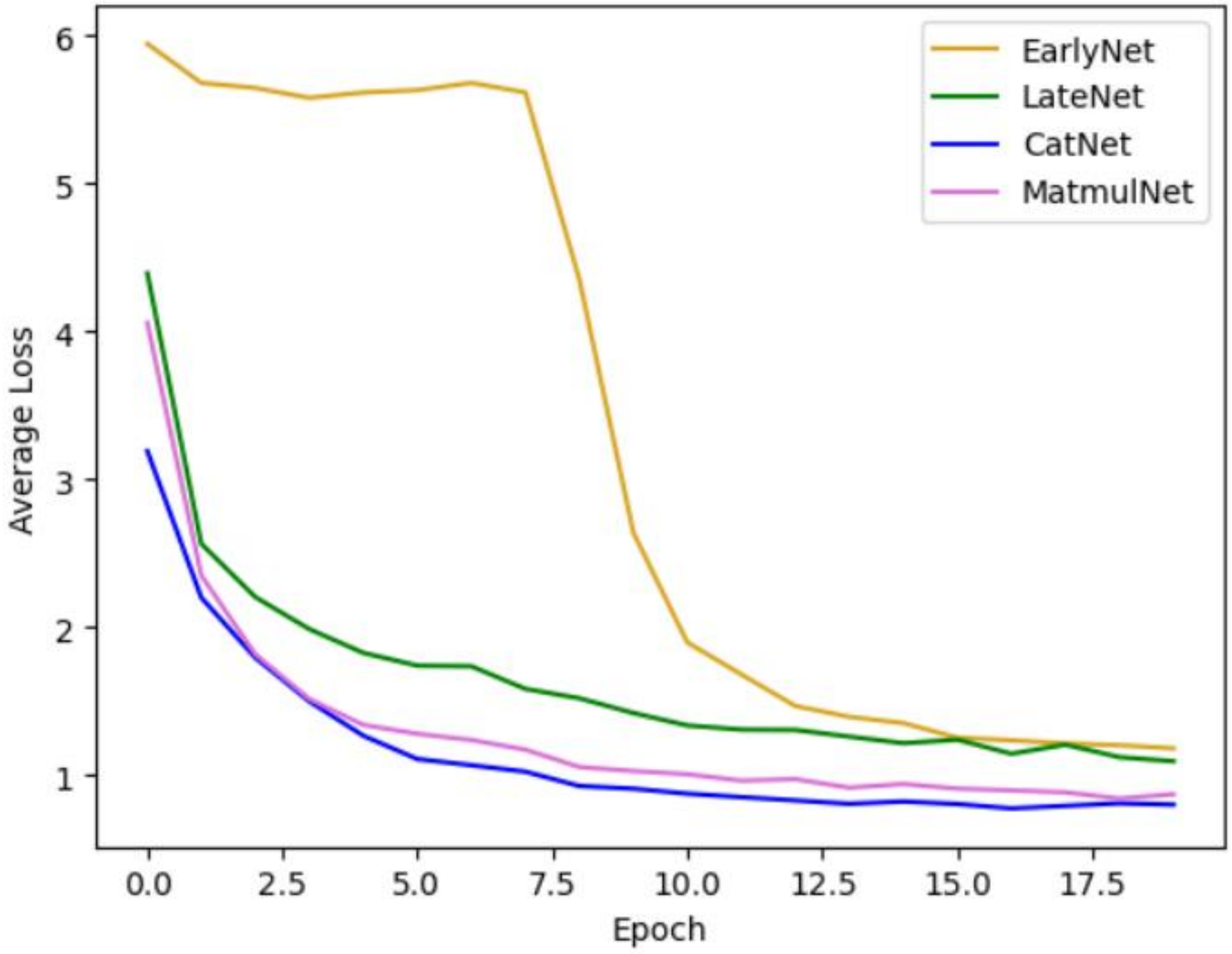
Part 3: Cross-modal Projection

Agenda

- Part 1: Multimodal Data
- Part 2: Intermediate Fusion
- Part 3: Cross-modal Projection and OCR Pipelines
- Part 4: VSS and GraphRAG
- Part 5: Wrap-up & Assessment

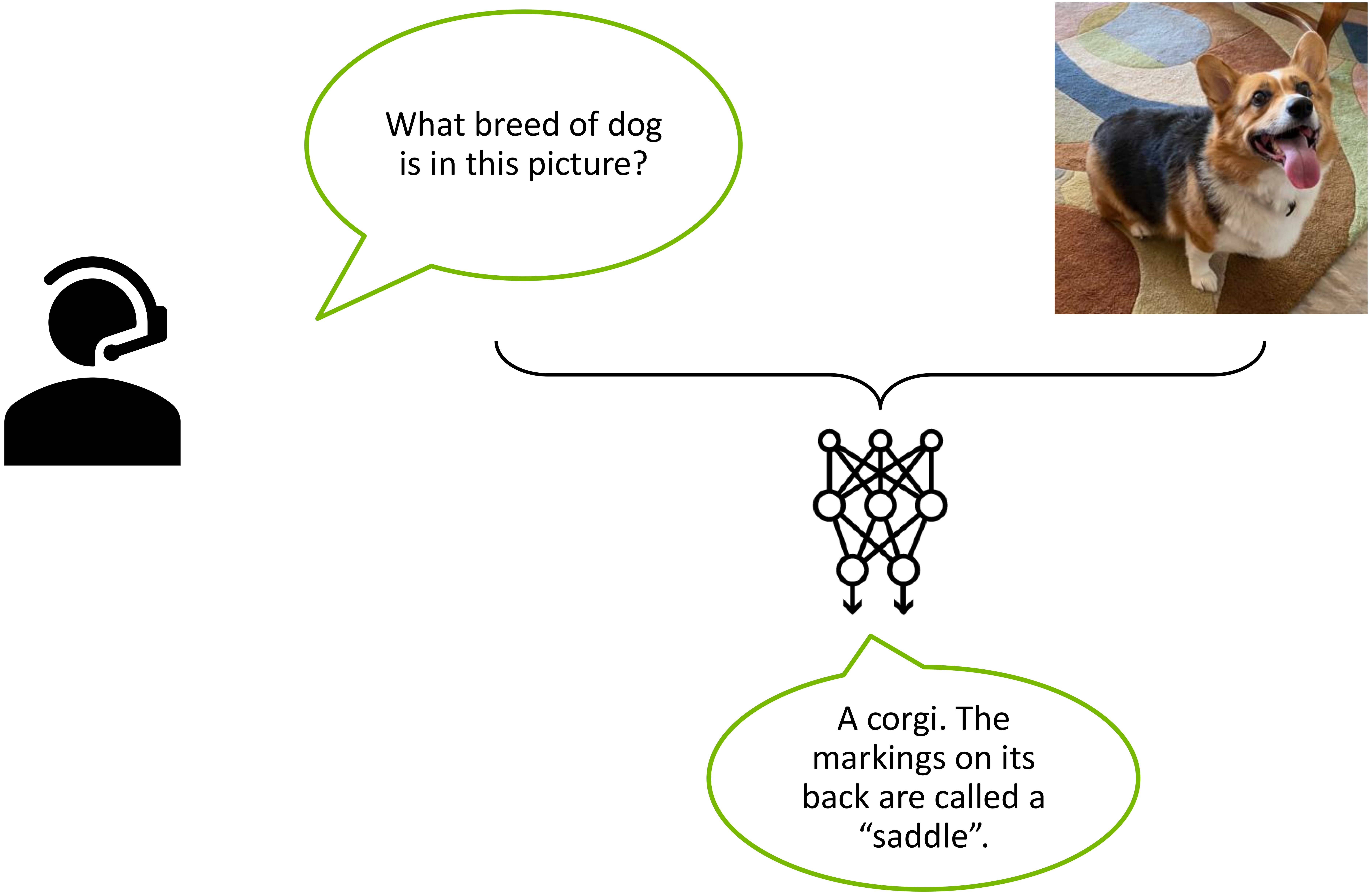
Experiment Results

Experiment Results



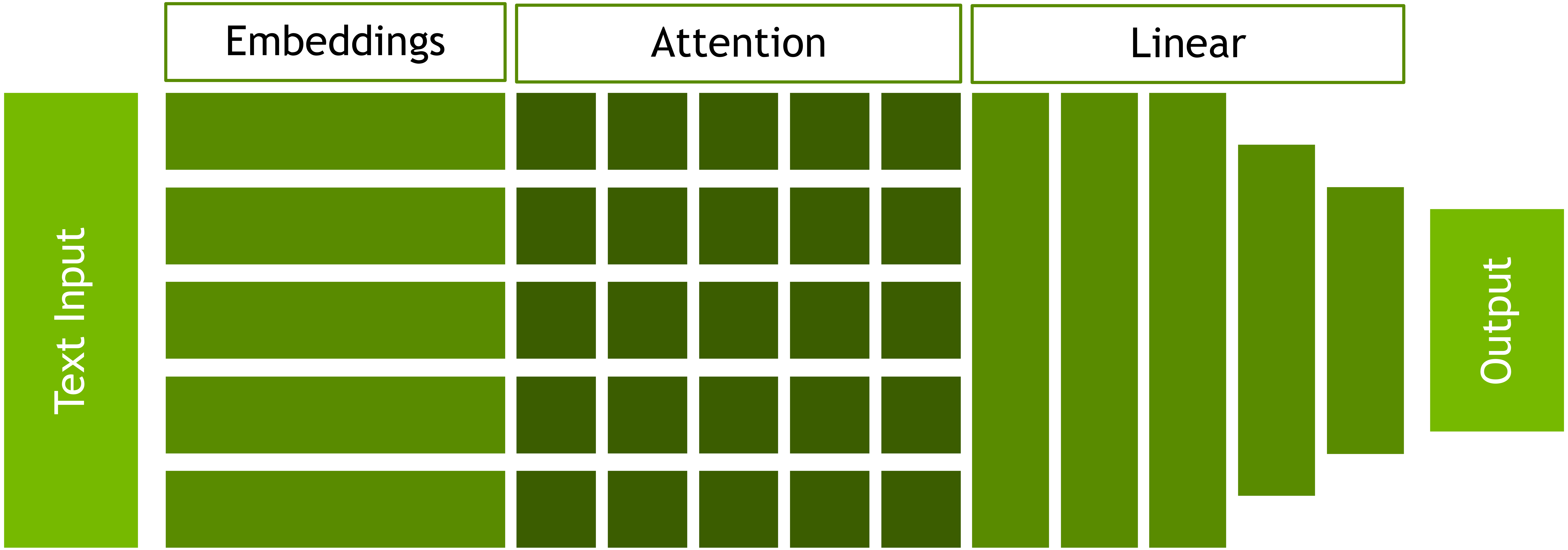
Cross-modal Projection

Vision Language Models



LLM Architecture

A simplified overview



LLM Architecture

A simplified overview

Visual Instruction Tuning

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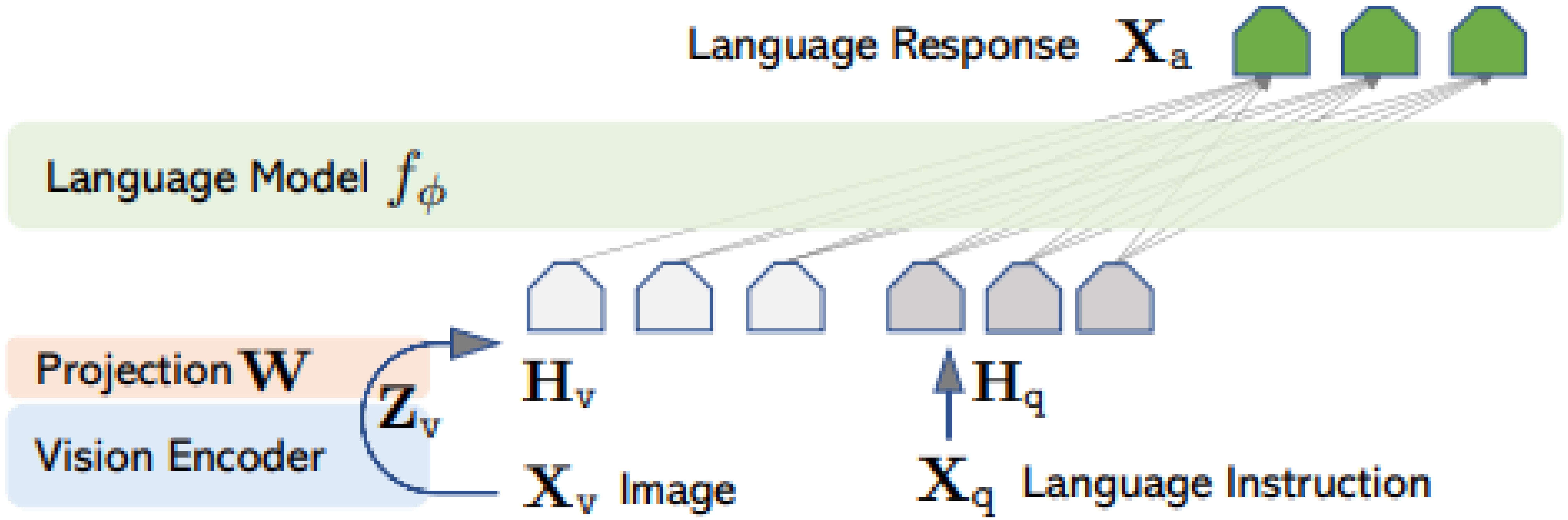
[1]University of Wisconsin–Madison [2]Microsoft Research [3]Columbia University
https://llava-vl.github.io

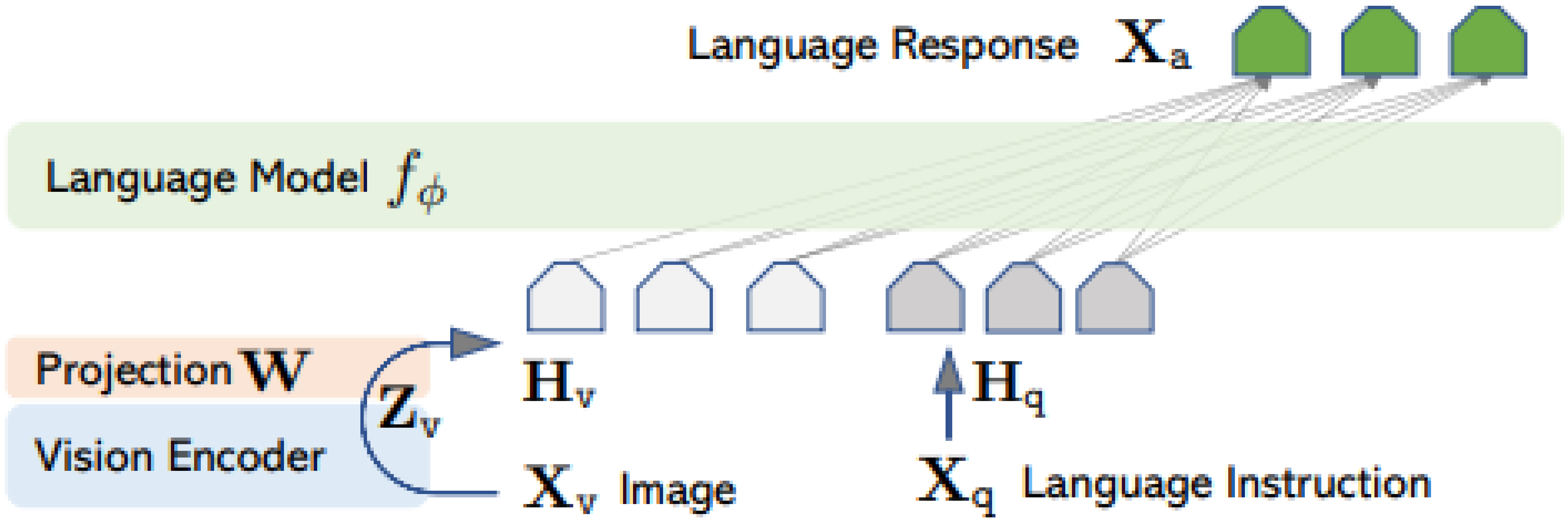
Abstract

Instruction tuning large language models (LLMs) using machine-generated instruction-following data has been shown to improve zero-shot capabilities on new tasks, but the idea is less explored in the multimodal field. We present the first attempt to use language-only GPT-4 to generate multimodal language-image instruction-following data. By instruction tuning on such generated data, we introduce LLaVA: **L**arge **L**anguage **a**nd **V**ision **A**ssistant, an end-to-end trained large multimodal model that connects a vision encoder and an LLM for general-purpose visual and language understanding. To facilitate future research on visual instruction following, we construct two evaluation benchmarks with diverse and challenging application-oriented tasks. Our experiments show that LLaVA demonstrates impressive multimodal chat abilities, sometimes exhibiting the behaviors of multimodal GPT-4 on unseen images/instructions, and yields a 85.1% relative score compared with GPT-4 on a synthetic multimodal instruction-following dataset. When fine-tuned on Science QA, the synergy of LLaVA and GPT-4

The Projection Matrix



The Projection Matrix



$$H_v = W \cdot Z_v, with\ Z_v = g(X_v)$$

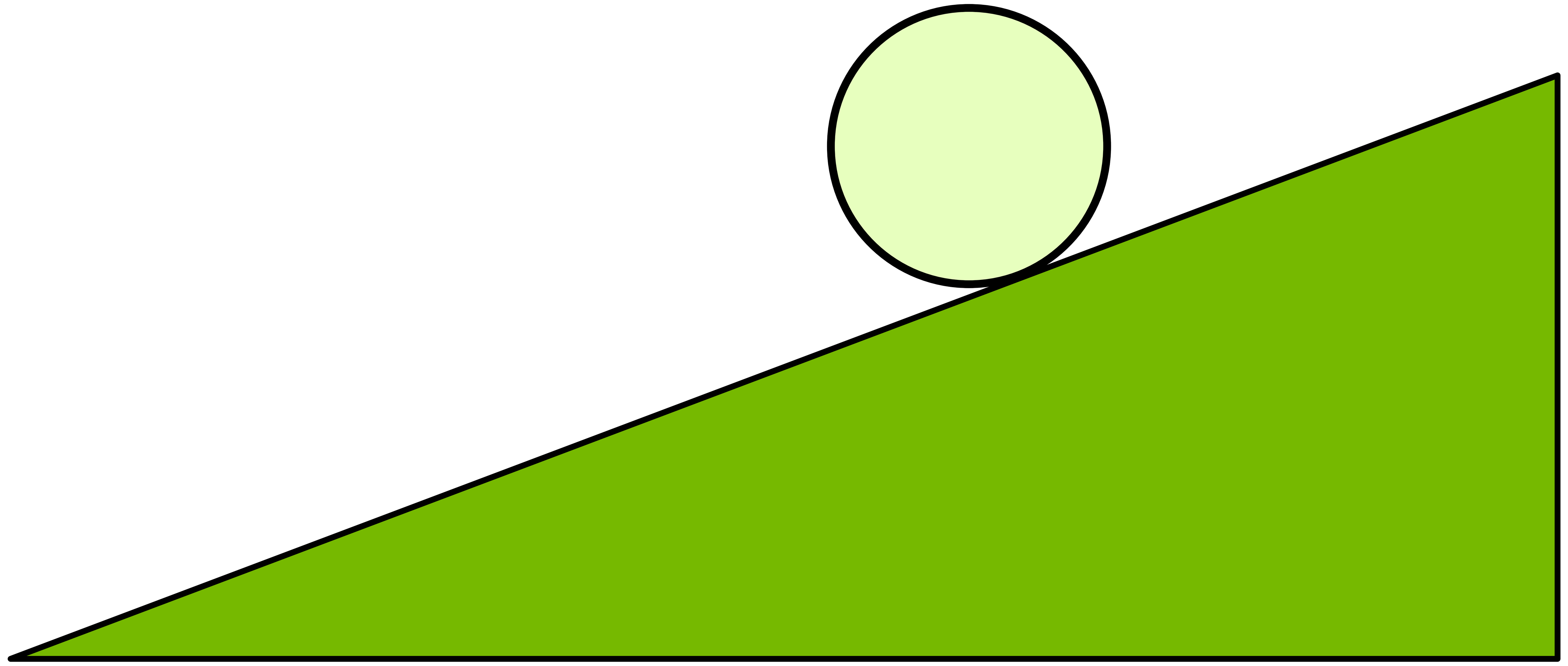
Projection in Physics

The Projection Matrix

Marble Physics

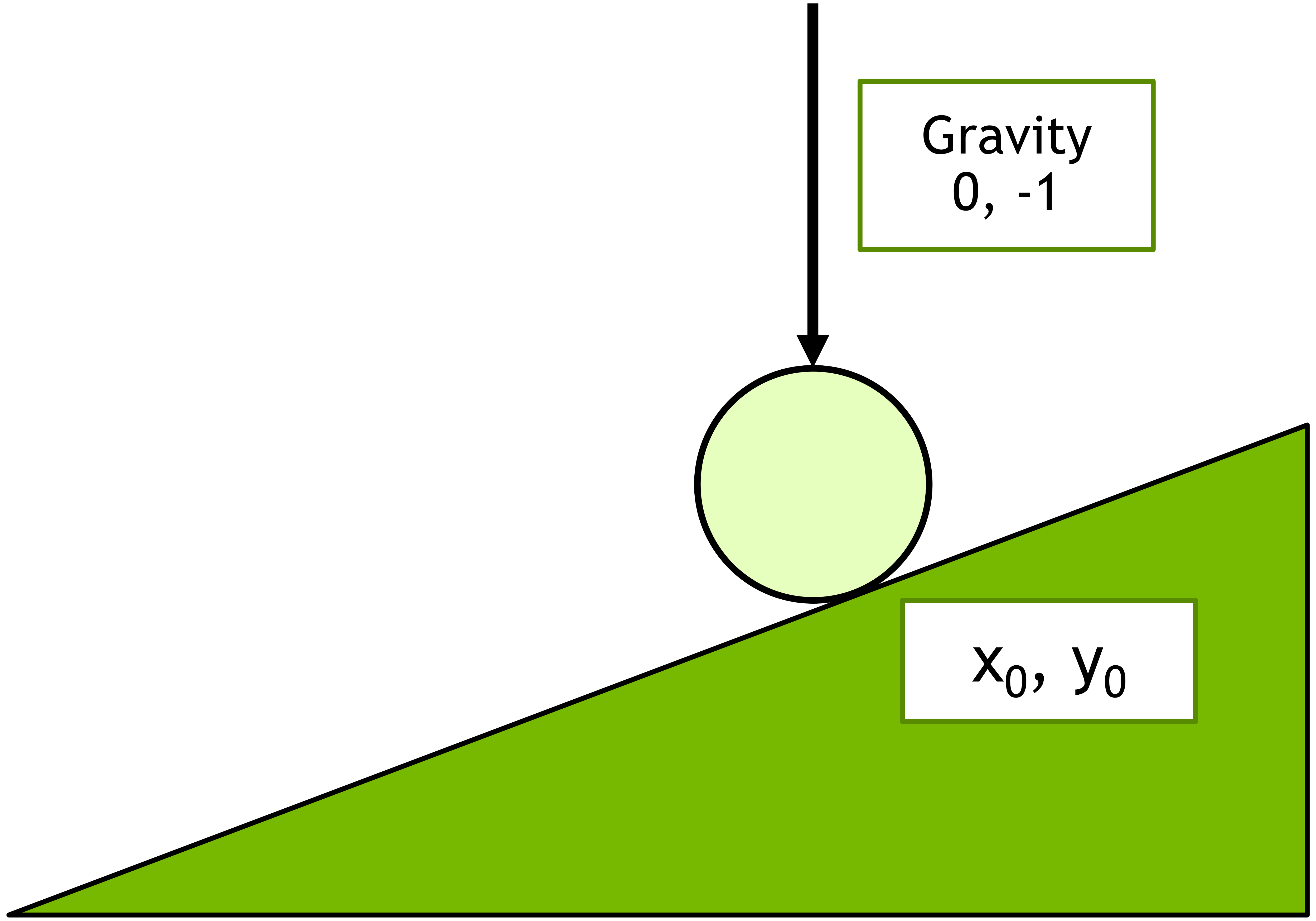
The Projection Matrix

Marble Physics



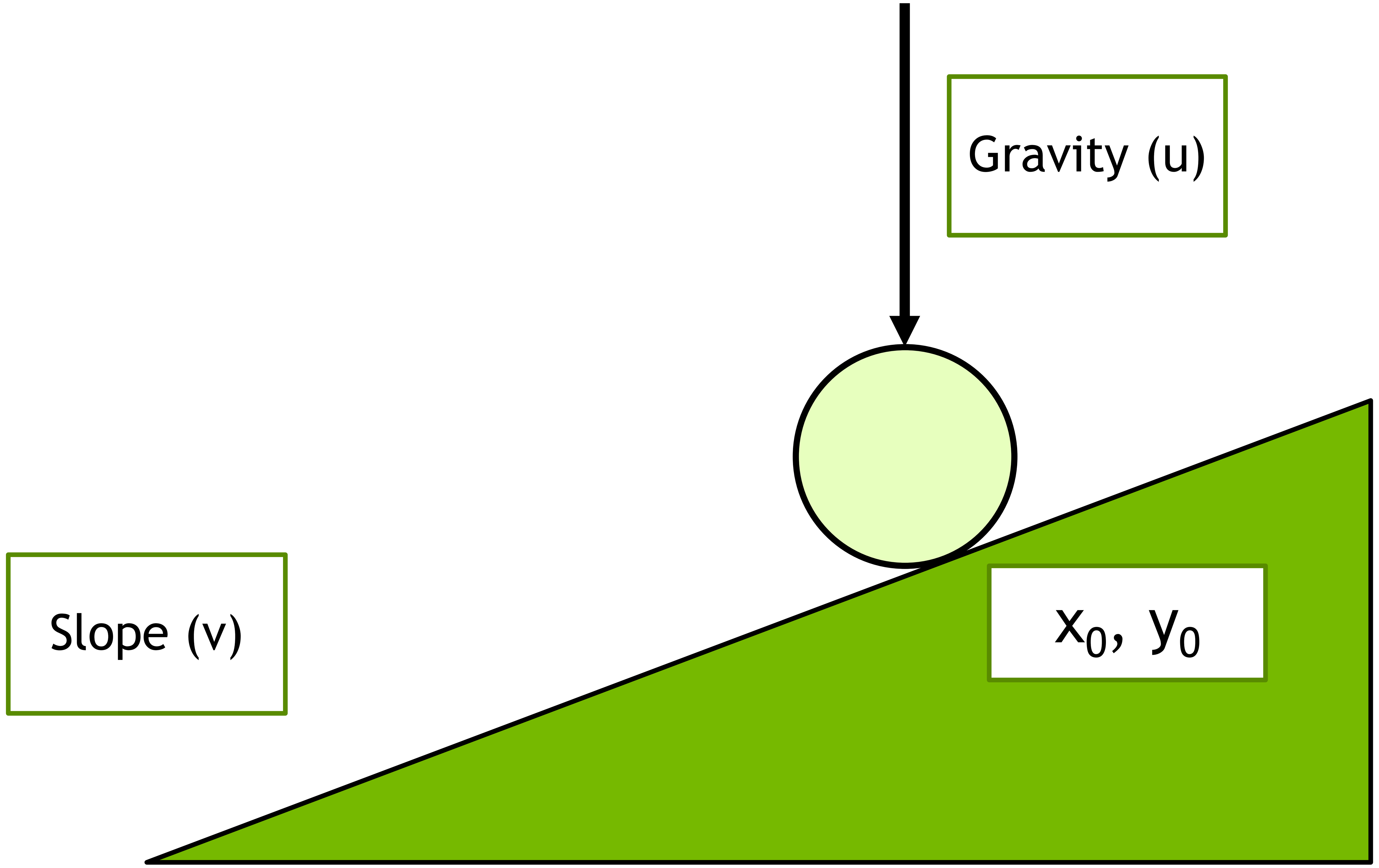
The Projection Matrix

Marble Physics



The Projection Matrix

Marble Physics



The Projection Matrix

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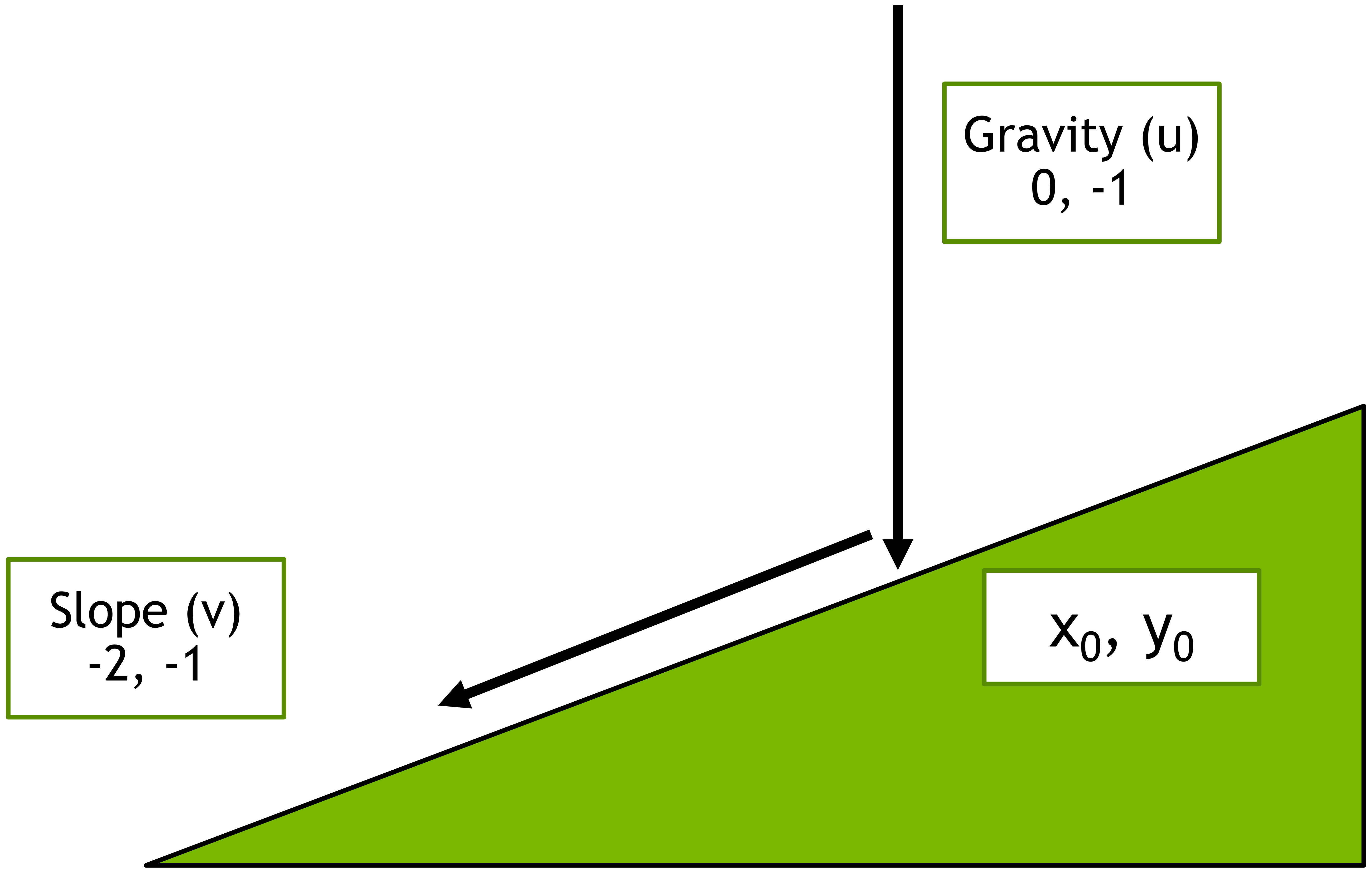

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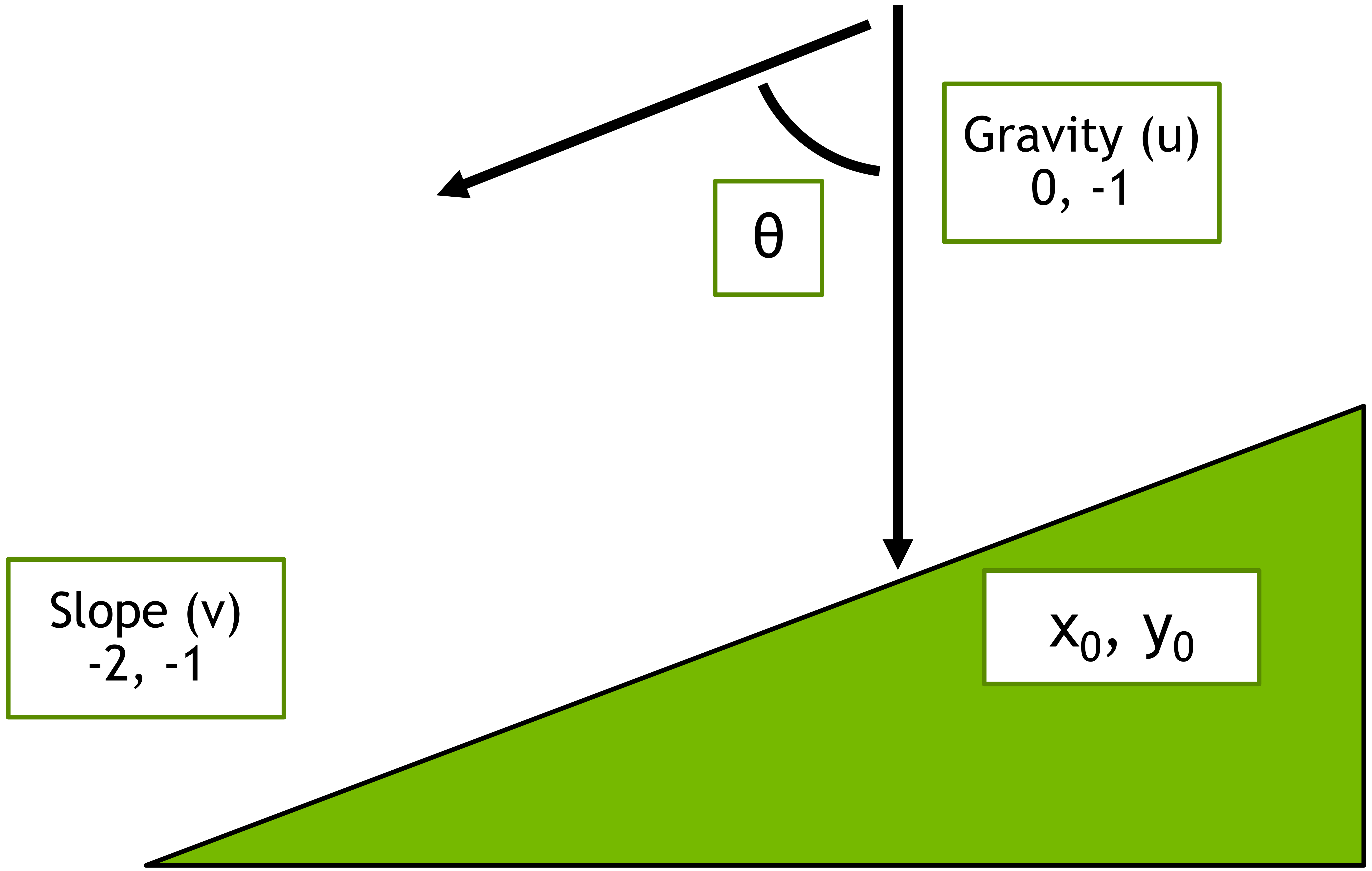
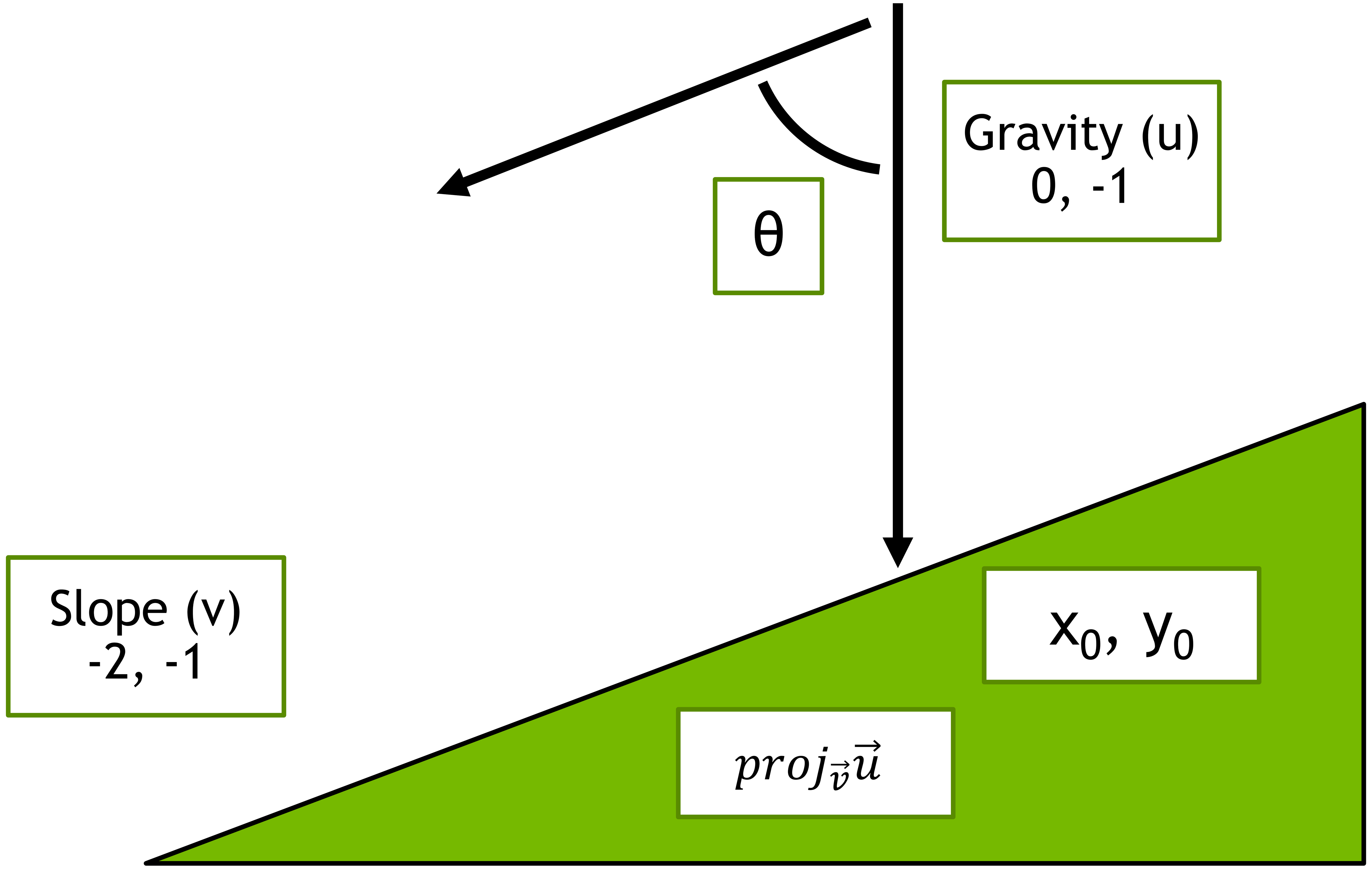


The Projection Matrix

Marble Physics

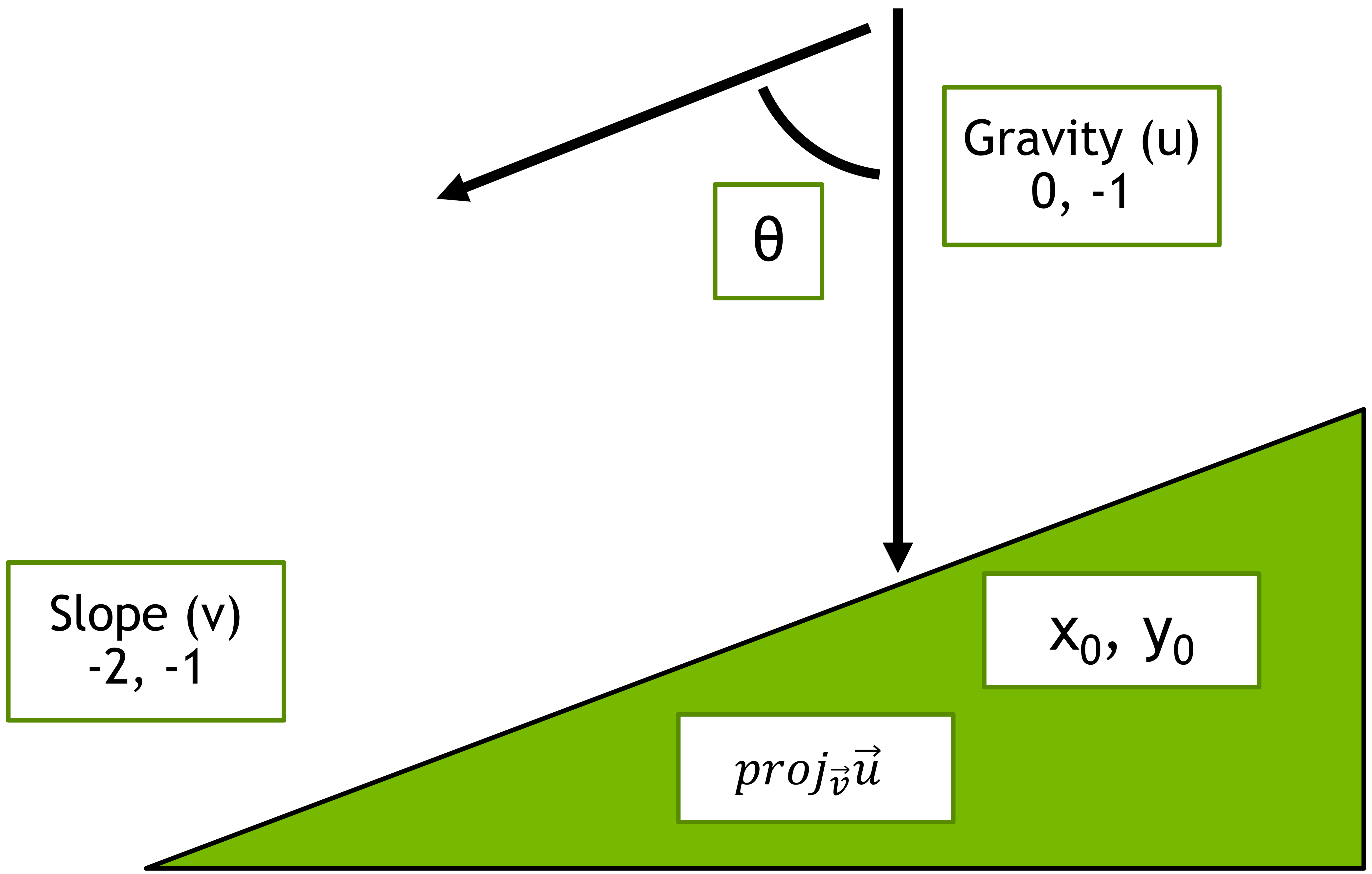


$$proj_{\vec{v}}\vec{u} = \|\vec{u}\| \times \frac{\vec{v}}{\|\vec{v}\|} \times cos(\theta)$$

$$\|\vec{u}\| = length\ of\ \vec{u}$$

The Projection Matrix

Marble Physics



$$proj_{\vec{v}}\vec{u} = \|\vec{u}\| \times \frac{\vec{v}}{\|\vec{v}\|} \times cos(\theta)$$

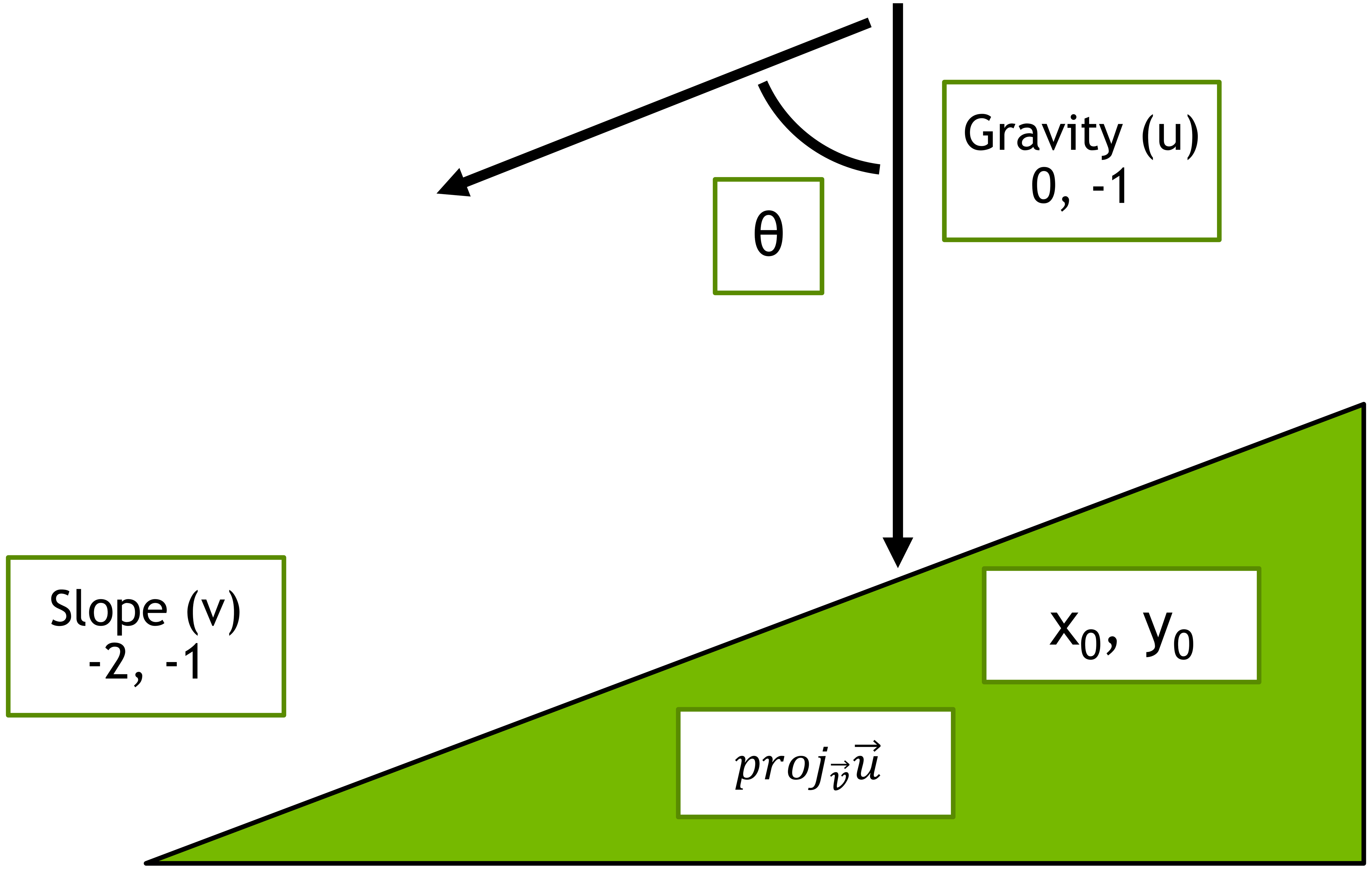
$$\theta = 63^{\circ}$$

$$proj_{\vec{v}}\vec{u} = 1 \times \frac{\langle -2, -1 \rangle}{2.23} \times cos\left(63^{\circ}\right)$$

$$proj_{\vec{v}}\vec{u} = \langle -.41, -.20 \rangle$$

The Projection Matrix

Marble Physics



$$proj_{\vec{v}}\vec{u} = \|\vec{u}\| \times \boxed{\frac{\vec{v}}{\|\vec{v}\|}} \times cos(\theta)$$

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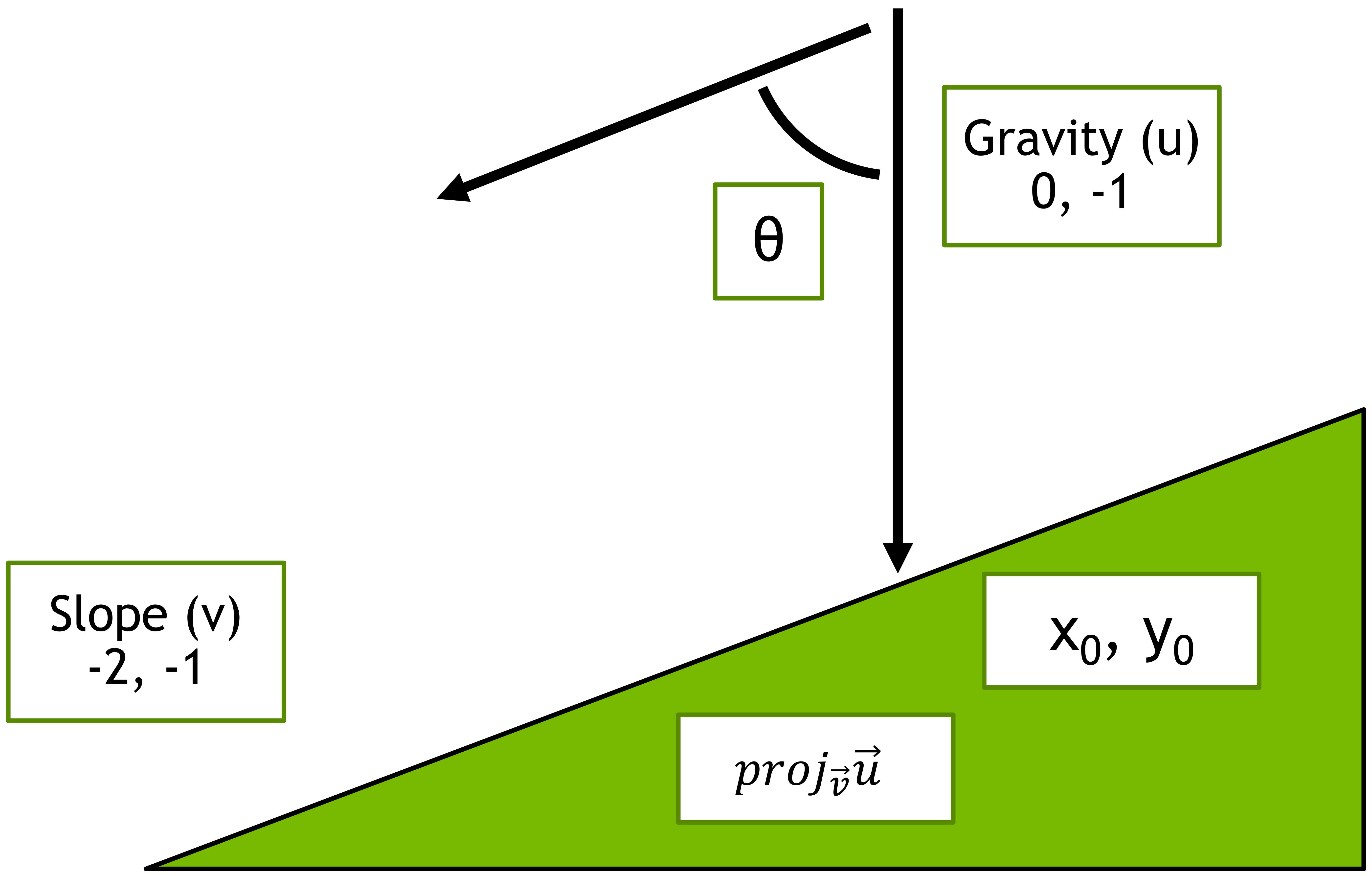

$$proj_{\vec{v}}\vec{u} = \|\vec{u}\| \times \frac{\vec{v}}{\|\vec{v}\|} \times cos(\theta)$$

$$\theta = 63^{\circ}$$

$$proj_{\vec{v}}\vec{u} = 1 \times \frac{\langle -2,-1\rangle}{2.23} \boxed{\times\, cos\left(63^{\circ}\right)}$$

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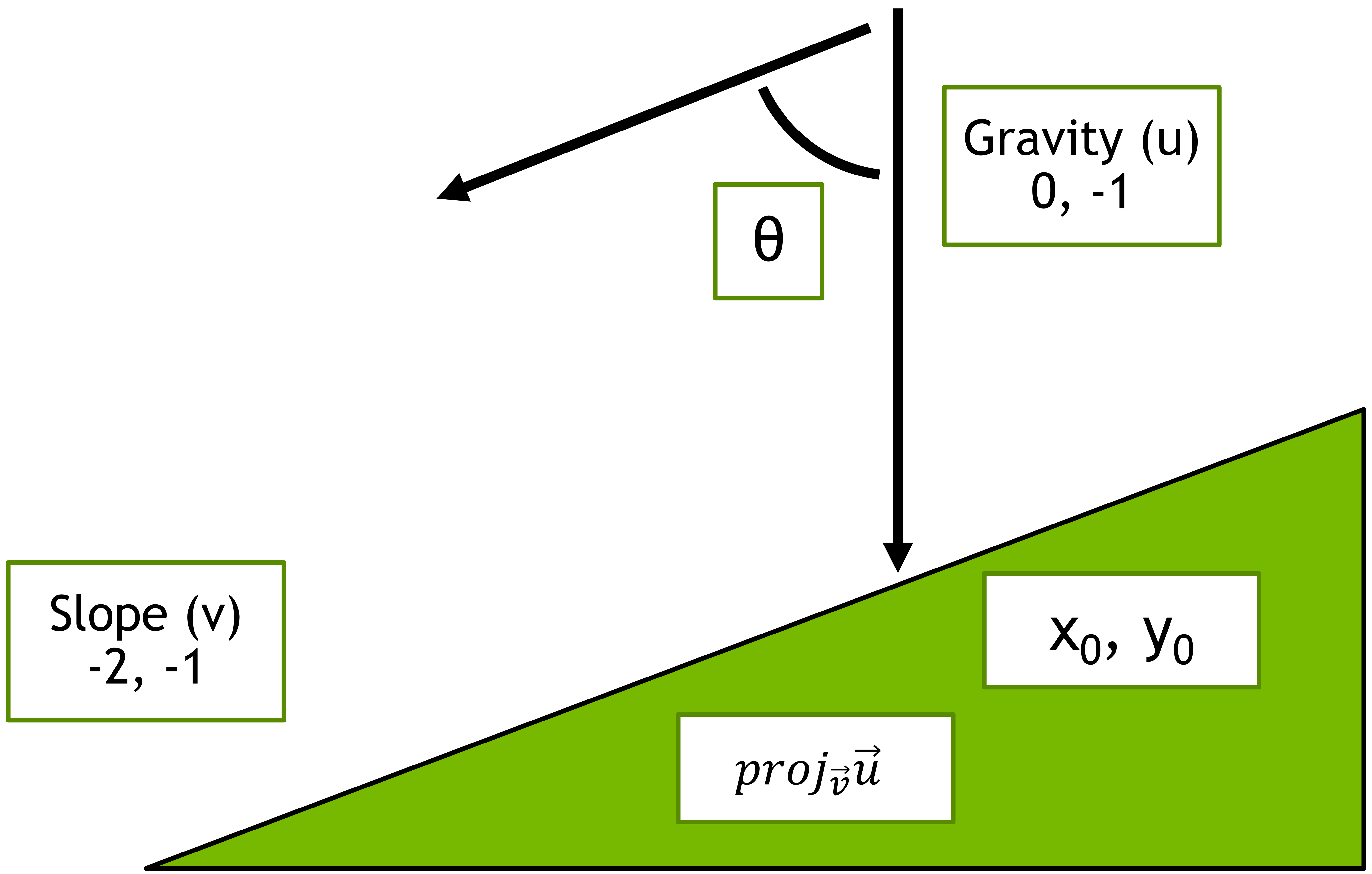
The Projection Matrix

Marble Physics



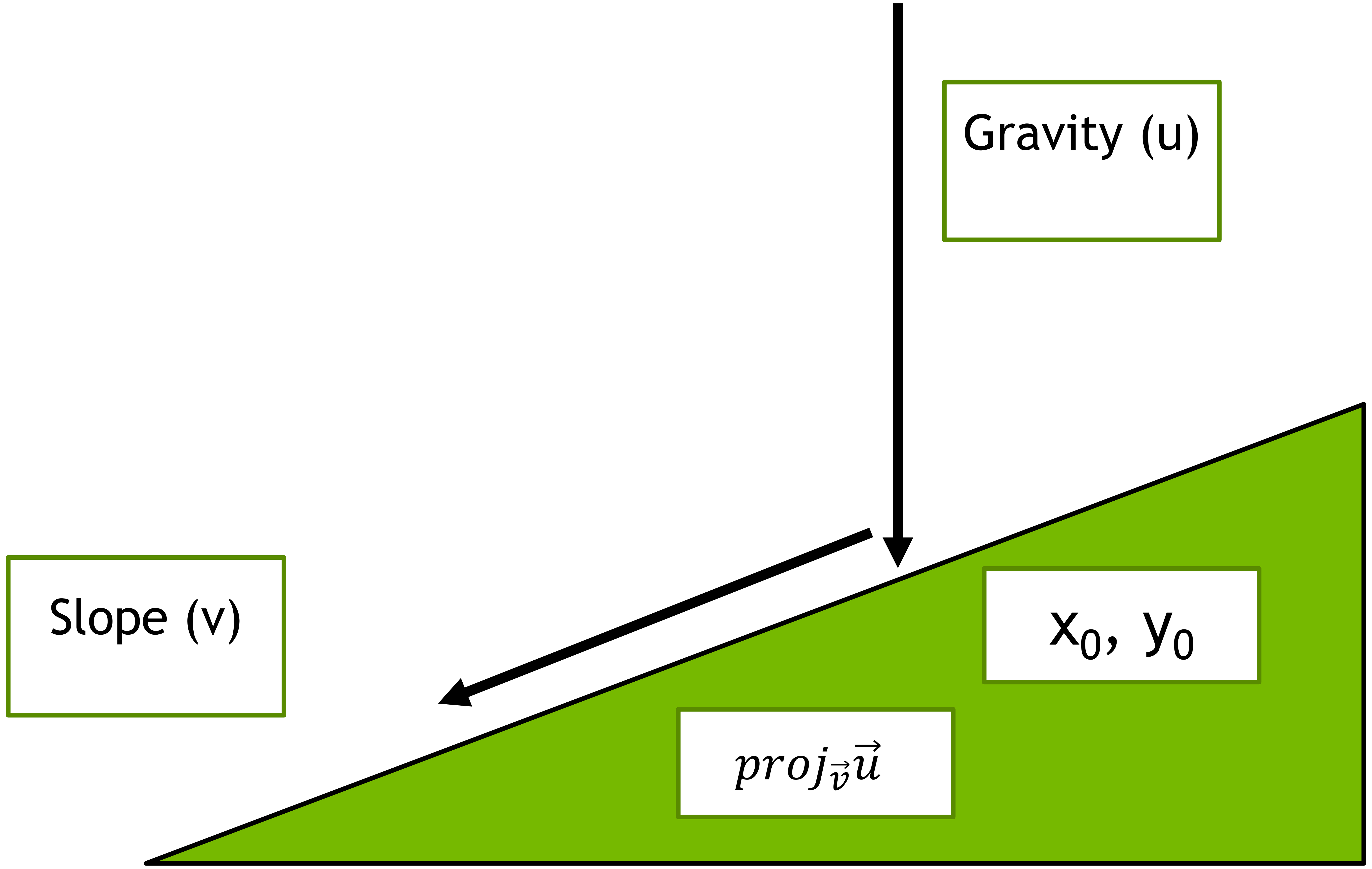
$$proj_{\vec{v}}\vec{u} = \|\vec{u}\| \times \frac{\vec{v}}{\|\vec{v}\|} \times cos(\theta)$$

$$proj_{\vec{v}}\vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}\right)\vec{v}$$

Projection in Visual Instruction Tuning

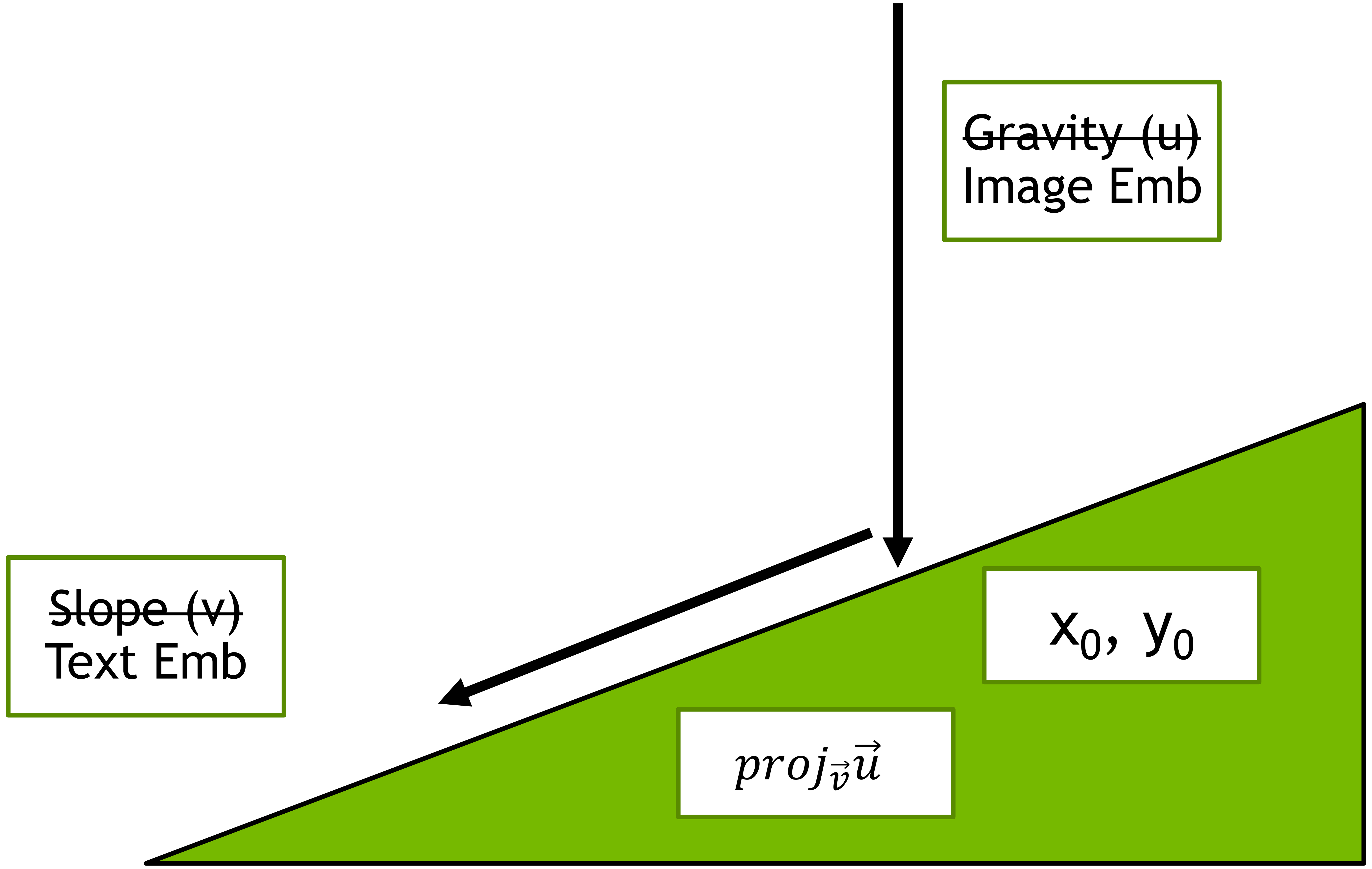
The Projection Matrix

Marble Physics



The Projection Matrix

Marble Physics



The Projection Matrix

Matrix Multiplication

x_1	x_2	x_3	x_4

w_{11}	w_{12}	w_{13}	w_{14}		$\hat{y}_1$
w_{21}	w_{22}	w_{23}	w_{24}		$\hat{y}_2$
w_{31}	w_{32}	w_{33}	w_{34}		$\hat{y}_3$

$$\hat{y} = Wx$$

The Projection Matrix

Matrix Multiplication

x_1	x_2	x_3	x_4

$$\hat{y} = Wx$$

w_{11}	w_{12}	w_{13}	w_{14}		$\hat{y}_1$
w_{21}	w_{22}	w_{23}	w_{24}		$\hat{y}_2$
w_{31}	w_{32}	w_{33}	w_{34}		$\hat{y}_3$

$$= x_1 w_{11} + x_2 w_{12} + x_3 w_{13} + x_4 w_{14}$$

The Projection Matrix

Matrix Multiplication

x_1	x_2	x_3	x_4

w_{11}	w_{12}	w_{13}	w_{14}
w_{21}	w_{22}	w_{23}	w_{24}
w_{31}	w_{32}	w_{33}	w_{34}

$\hat{y}_1$
$\hat{y}_2$
$\hat{y}_3$

$$\hat{y} = Wx$$

$$= x_1 w_{11} + x_2 w_{12} + x_3 w_{13} + x_4 w_{14}$$

$$= x_1 w_{21} + x_2 w_{22} + x_3 w_{23} + x_4 w_{24}$$

The Projection Matrix

Matrix Multiplication

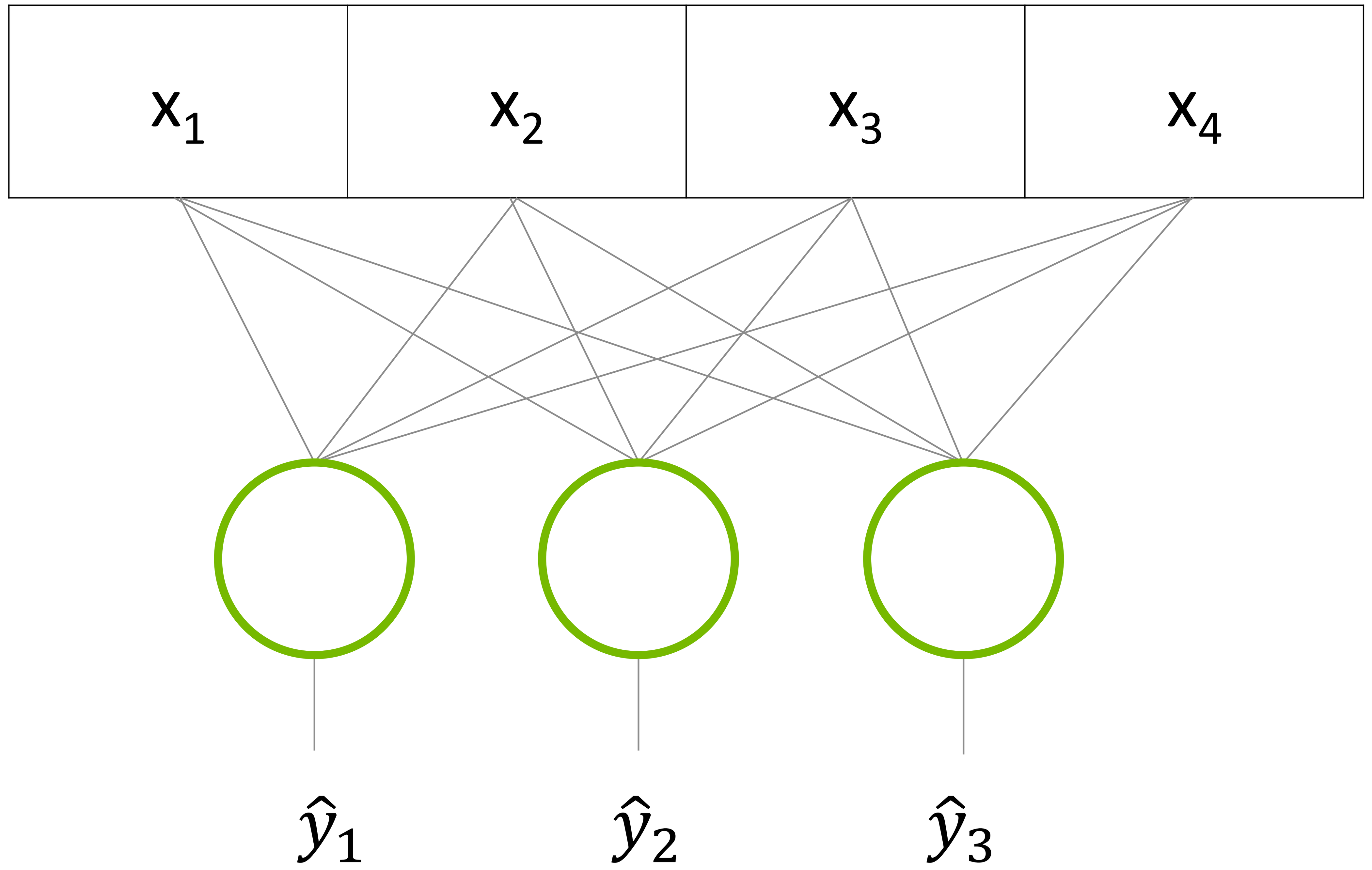
x_1	x_2	x_3	x_4

$$\hat{y} = Wx$$

w_{11}	w_{12}	w_{13}	w_{14}	$\hat{y}_1$	$= x_1 w_{11} + x_2 w_{12} + x_3 w_{13} + x_4 w_{14}$
w_{21}	w_{22}	w_{23}	w_{24}	$\hat{y}_2$	$= x_1 w_{21} + x_2 w_{22} + x_3 w_{23} + x_4 w_{24}$
w_{31}	w_{32}	w_{33}	w_{34}	$\hat{y}_3$	$= x_1 w_{31} + x_2 w_{32} + x_3 w_{33} + x_4 w_{34}$

The Projection Matrix

Matrix Multiplication

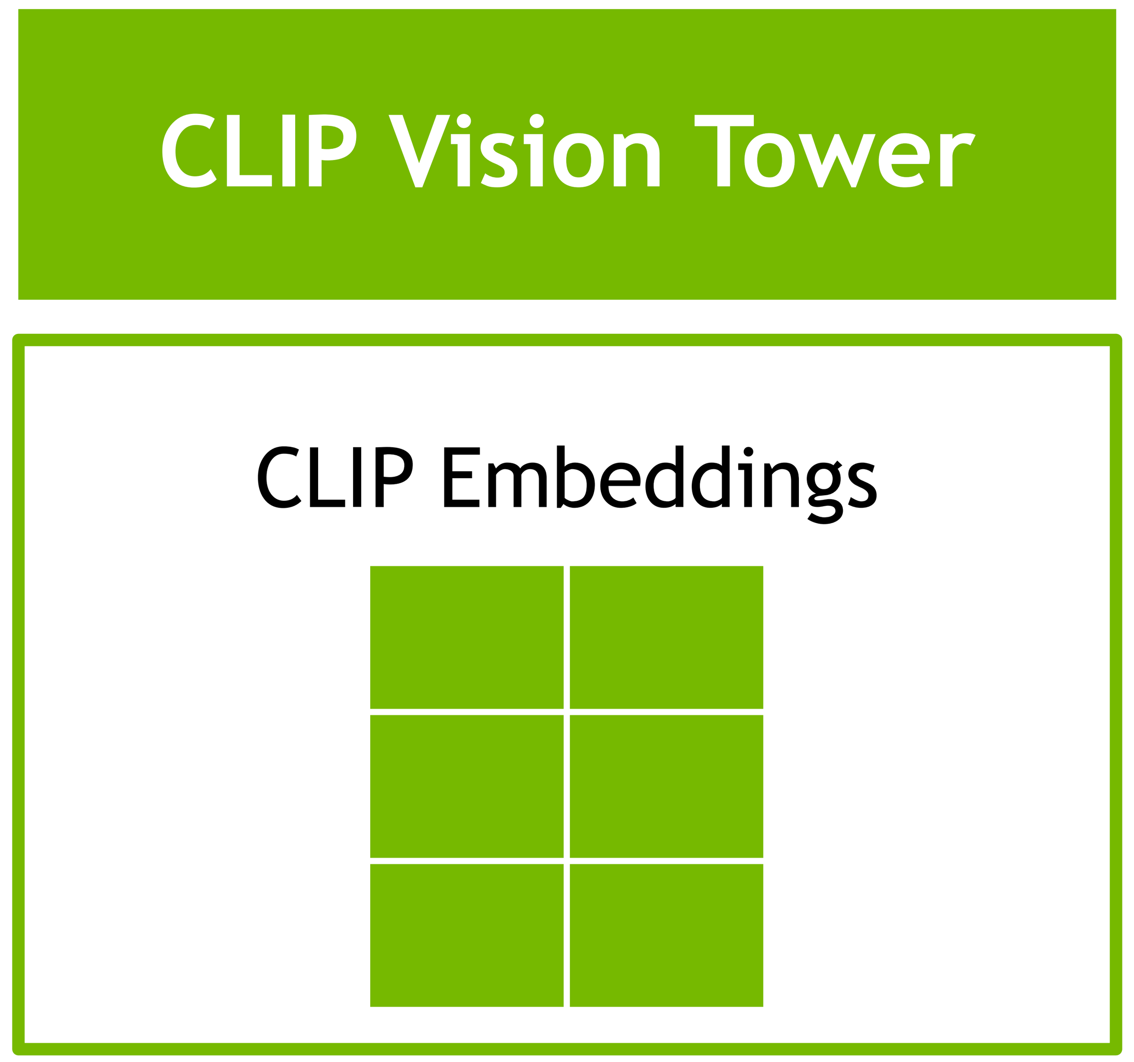


$$\hat{y} = Wx$$

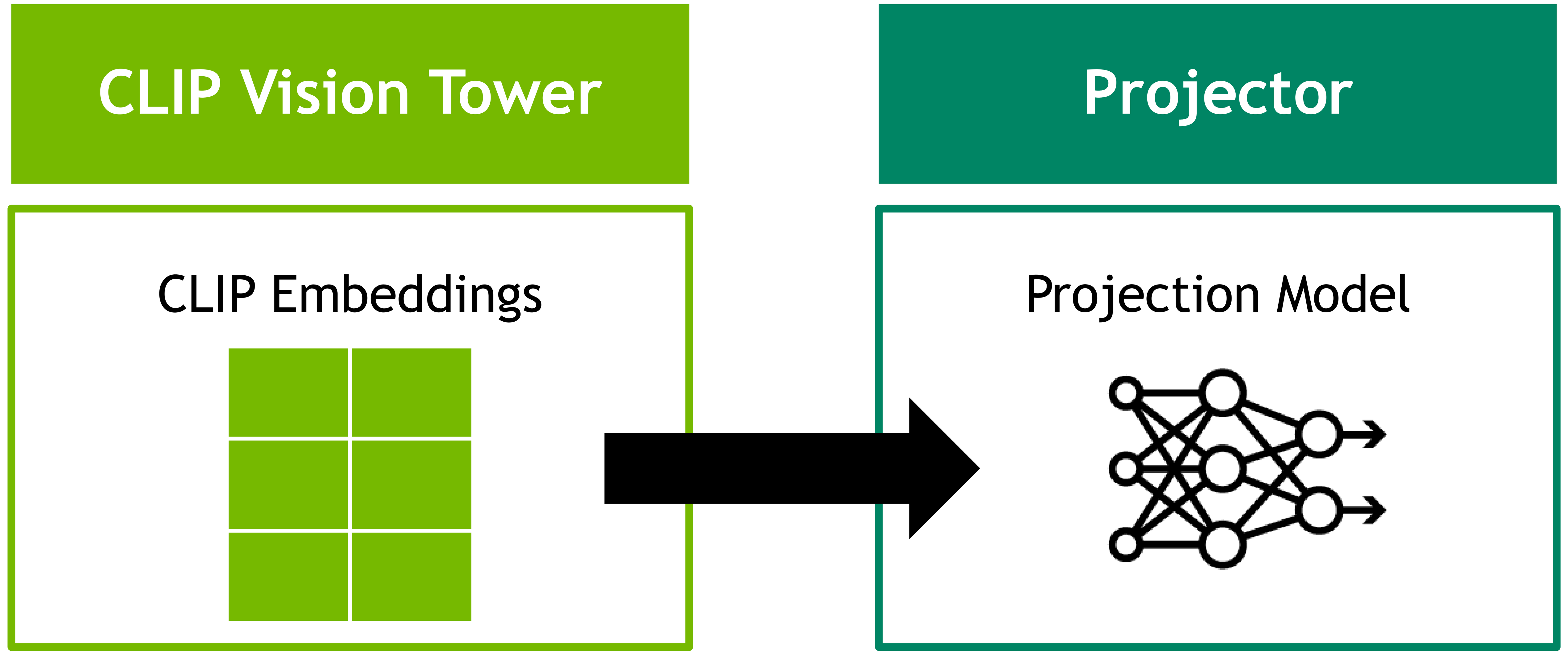
$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \end{bmatrix} \begin{aligned} &= x_1 w_{11} + x_2 w_{12} + x_3 w_{13} + x_4 w_{14} \\ &= x_1 w_{21} + x_2 w_{22} + x_3 w_{23} + x_4 w_{24} \\ &= x_1 w_{31} + x_2 w_{32} + x_3 w_{33} + x_4 w_{34} \end{aligned}$$

Visual Instruction Tuning in Practice

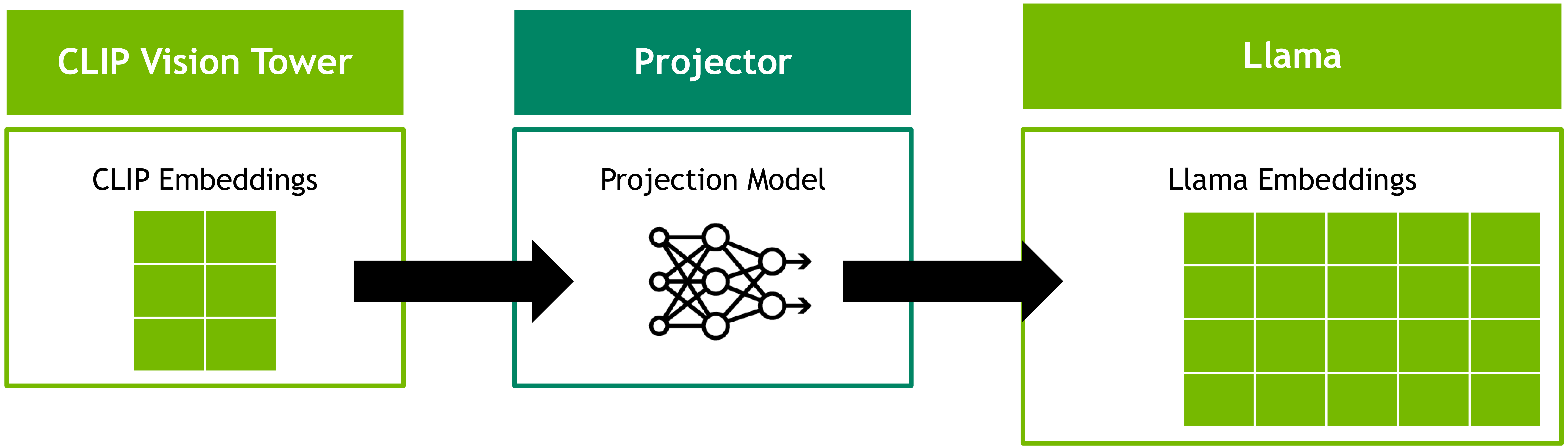
LlaVA Architecture



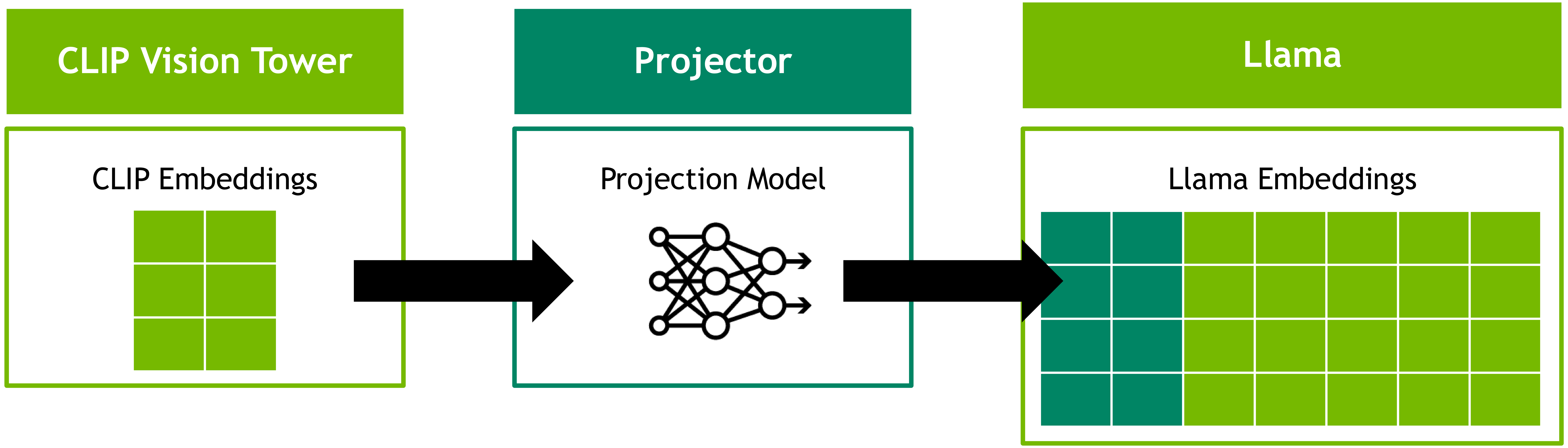
LlaVA Architecture



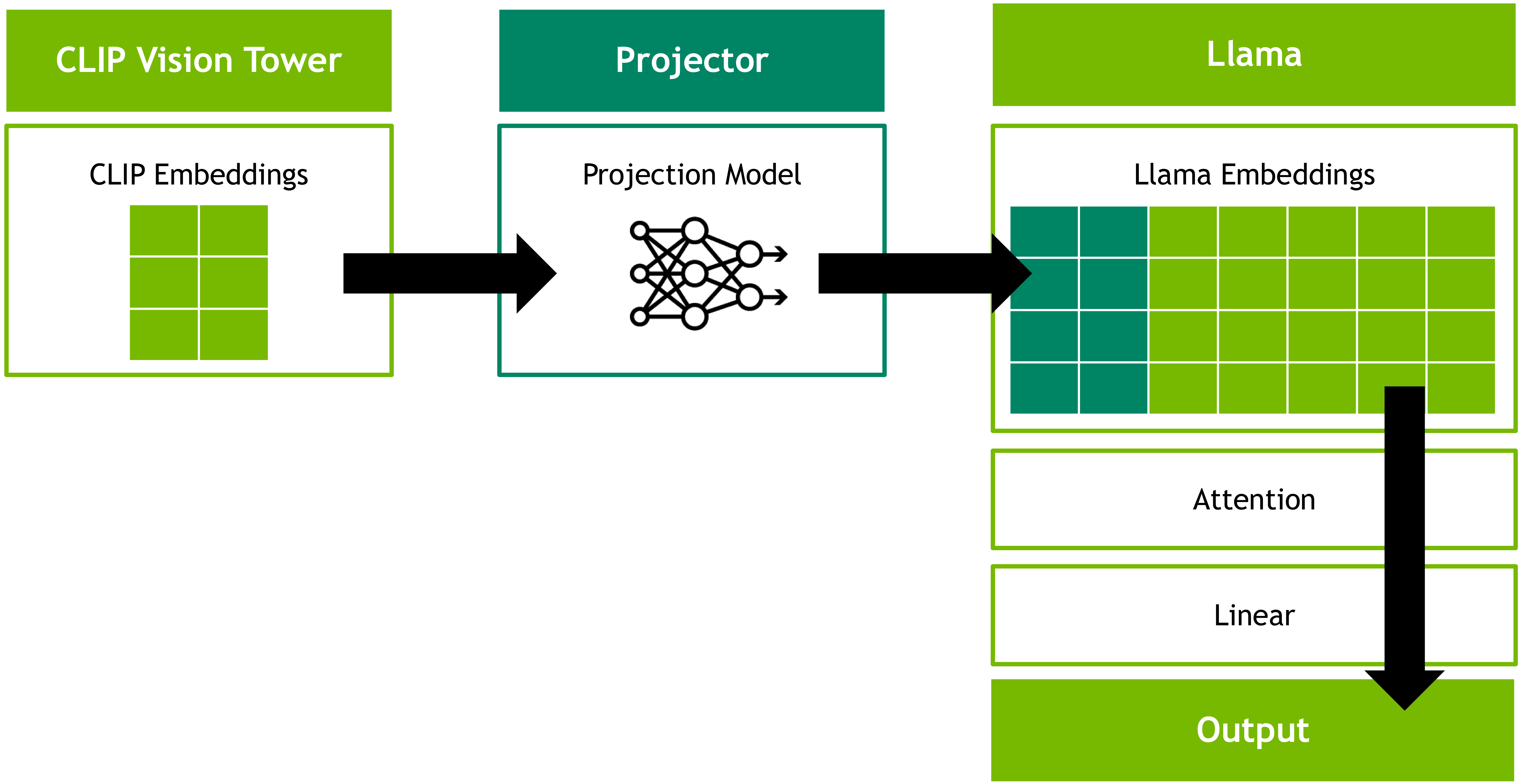
LlaVA Architecture



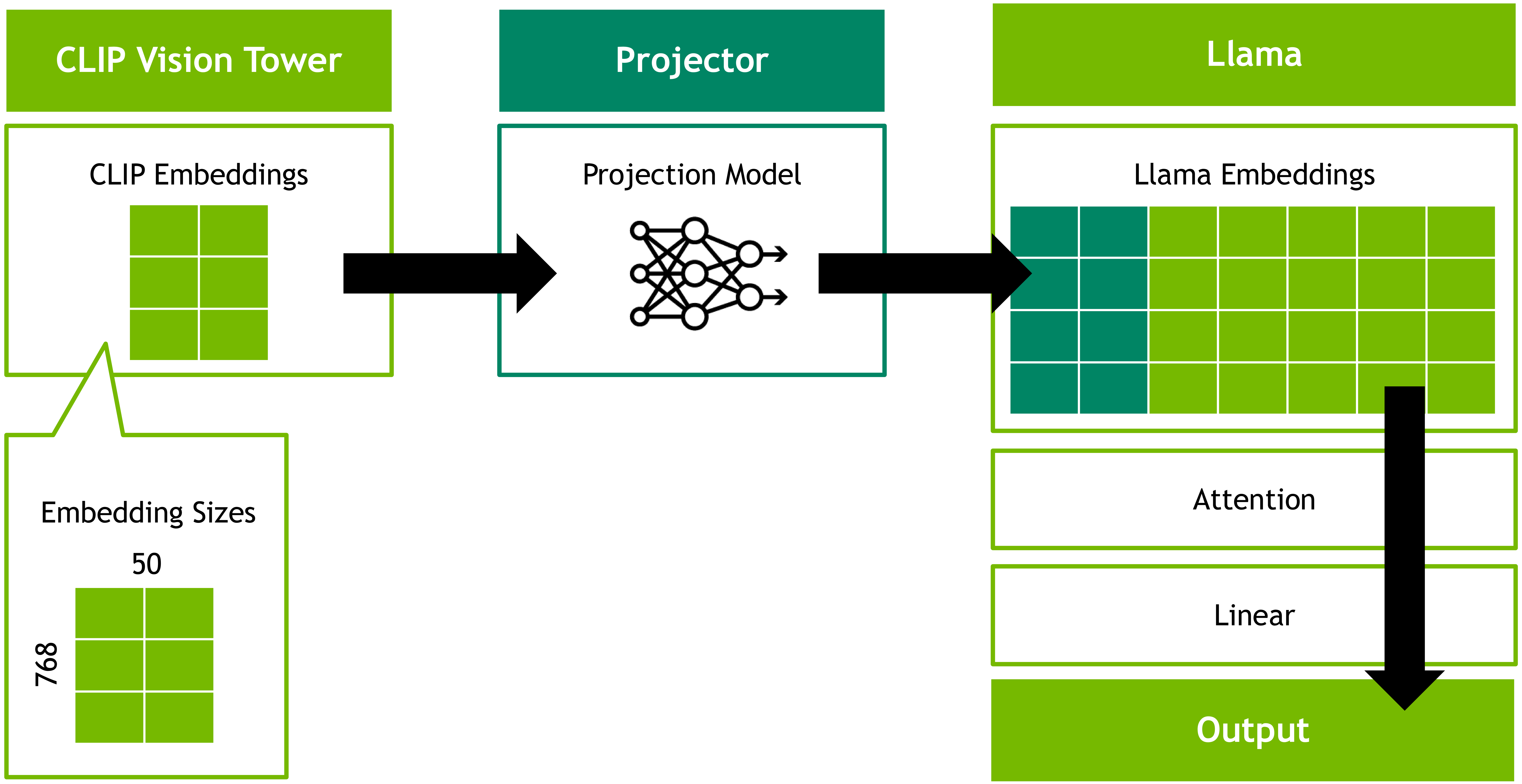
LlaVA Architecture



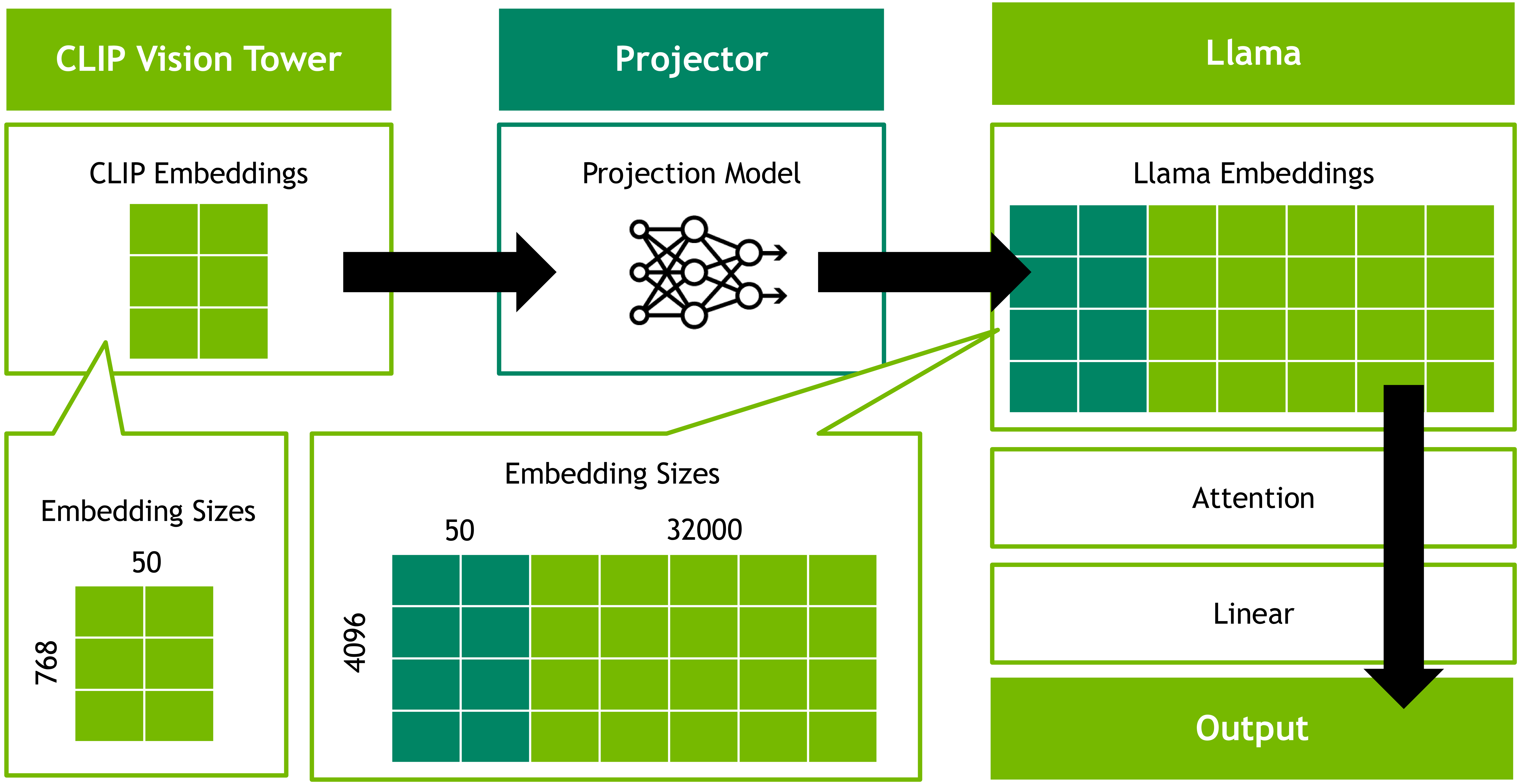
LlaVA Architecture



LlaVA Architecture

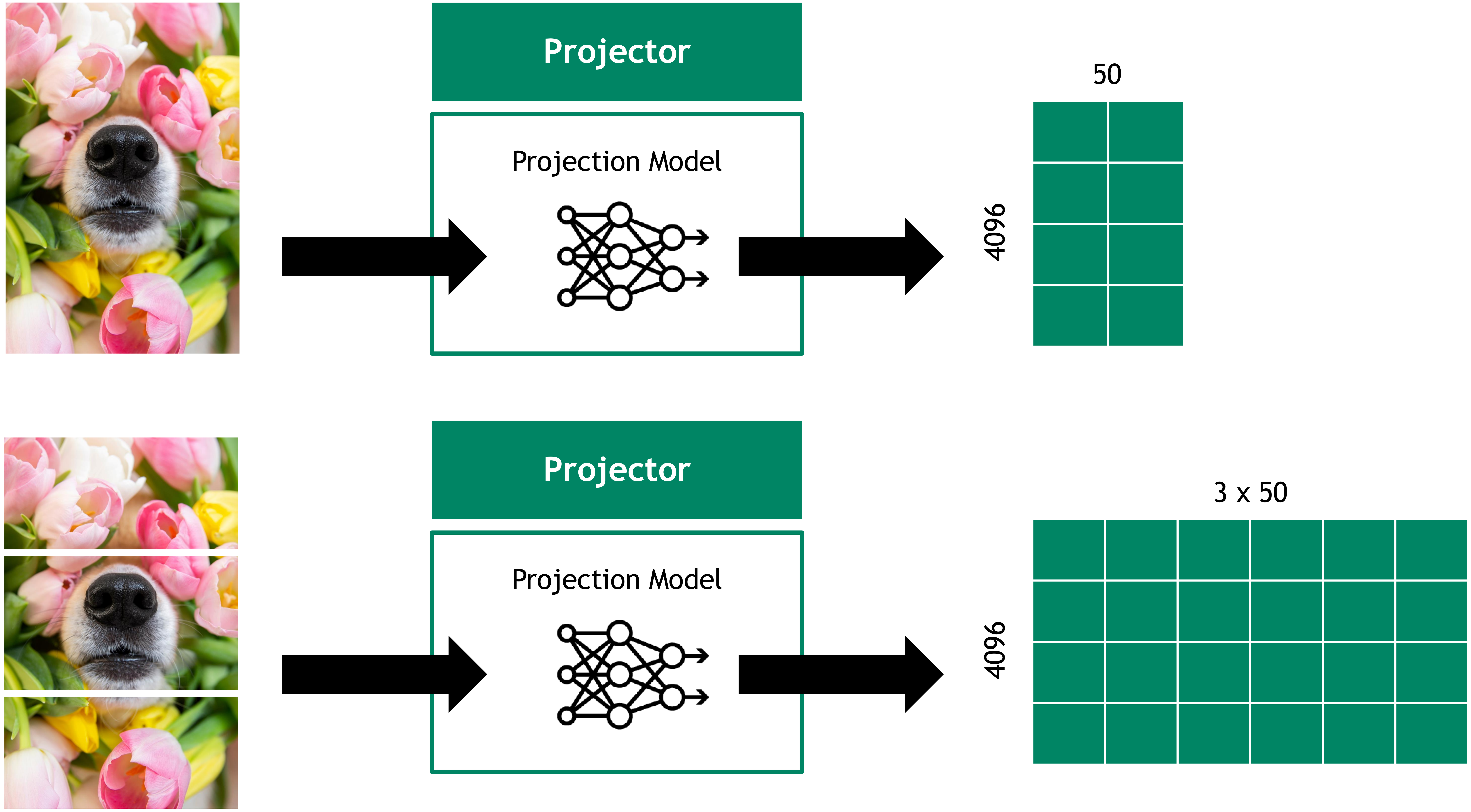


LlaVA Architecture



LlaVA Architecture

Patches



Lab 3a – Model Projection

“

Hypothesis:

If we can project image embeddings into text embeddings, we can project text embedding into image embeddings

”

Model Projection

Lab 3a

Convolutional Neural Network

Flatten

Linear

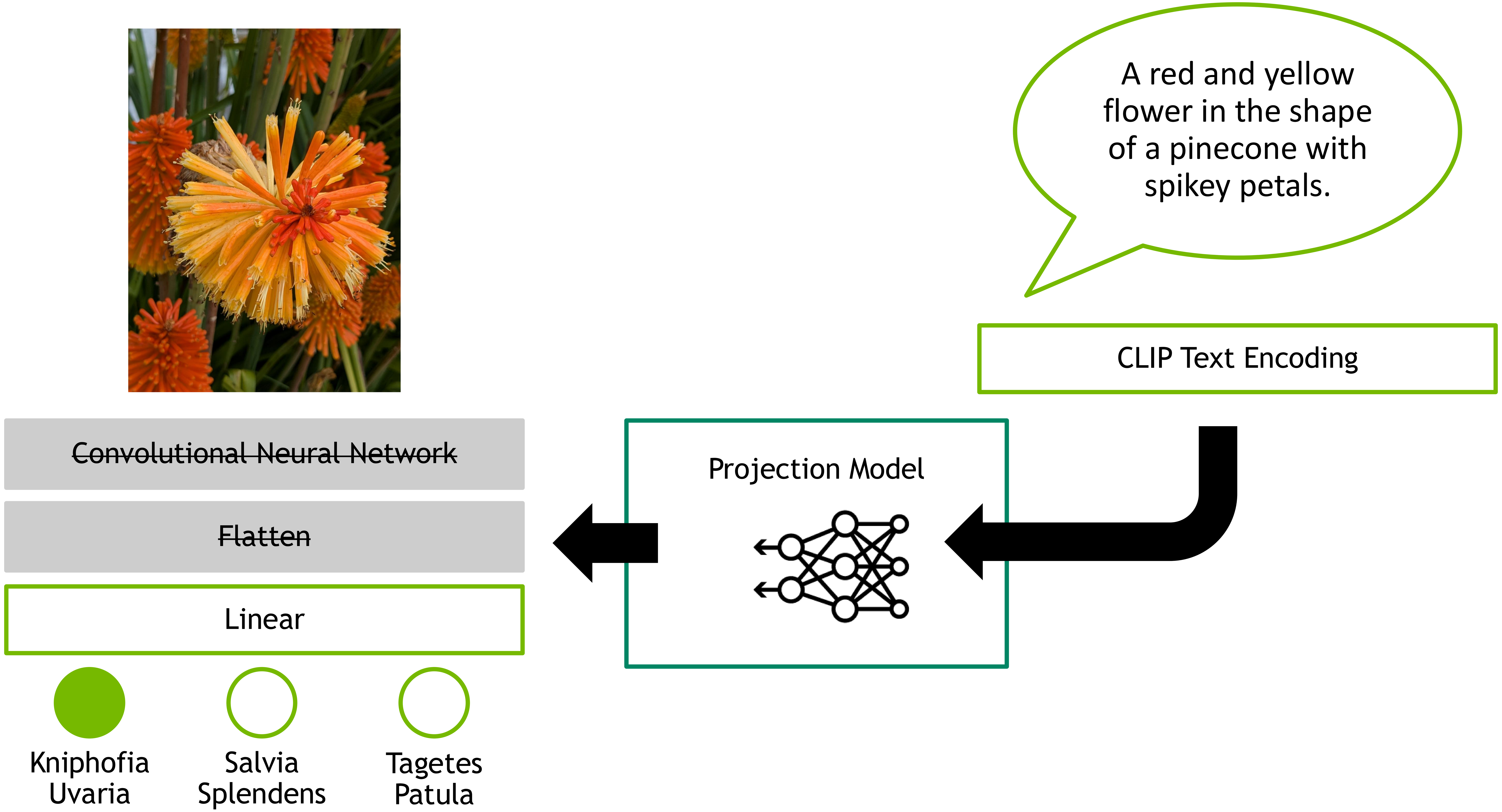
Kniphofia Uvaria

Salvia Splendens

Tagetes Patula

Model Projection

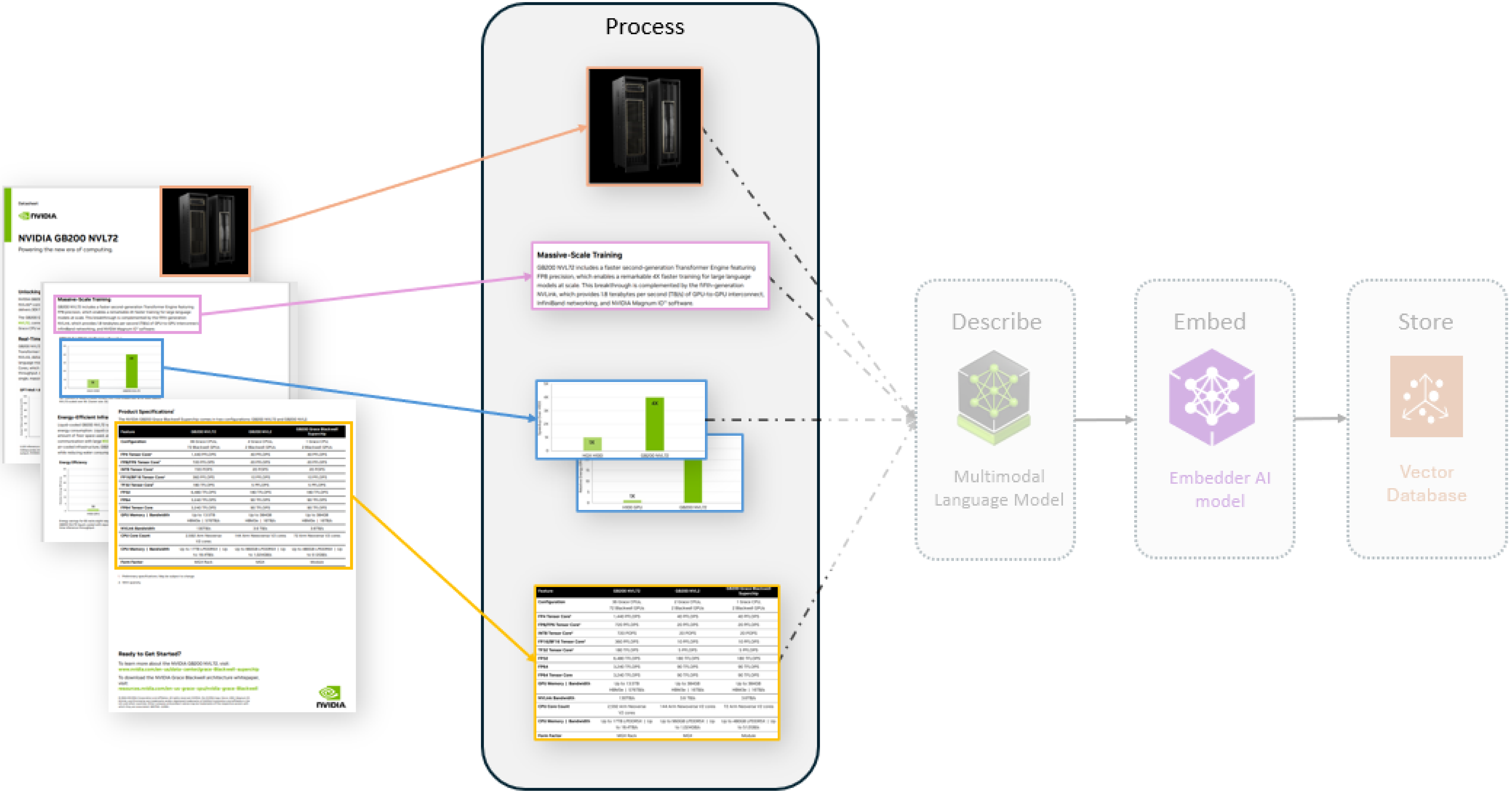
Lab 3a



Lab 3b – OCR Pipelines

OCR Pipelines

Lab 3b



Let’s Get Started!

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